

GNIBEN, Denmark

WMO: 061690

Lat: 56.0083N

Lon: 11.2783E

Elev: 39

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	23.5	25.6	16.5	12.7	25.9	19.4	14.6	27.5	40.5	37.5	36.5	36.7	13.9	140	0.768

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.7	74.4	67.4	71.9	66.1	69.8	64.6	68.8	72.7	67.0	70.8	65.4	69.1	9.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.3	100.4	71.4	65.4	93.9	69.6	63.7	88.5	68.0	33.0	72.7	31.5	70.9	30.3	69.2	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 37.0	33.2	30.2	22.0	79.2	3.6	2.6	19.4	81.1	17.3	82.6	15.3	84.1	12.7	86.0		
(5)			20.2	70.4	3.5	2.1	17.7	71.9	15.7	73.1	13.8	74.3	11.2	75.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.7	34.9	34.4	37.7	44.7	53.0	59.0	63.7	64.1	58.7	51.1	43.7	38.4		
	DBStd	11.50	4.85	4.72	4.34	4.45	4.52	4.07	4.22	3.95	3.71	4.13	4.47	5.28		
	HDD50	2067	468	437	382	169	20	0	0	0	0	37	193	361		
	HDD65	6036	933	857	847	610	373	184	78	66	191	431	640	826		
	CDD50	1589	0	0	0	9	112	271	425	436	262	71	3	0		
	CDD65	83	0	0	0	0	1	5	37	37	3	0	0	0		
	CDH74	89	0	0	0	0	1	3	45	39	1	0	0	0		
	CDH80	2	0	0	0	0	0	0	2	0	0	0	0	0		
	WSAvg	15.7	18.1	16.4	15.0	13.4	13.3	14.6	13.4	14.8	16.1	17.0	17.8	18.5		
	PrecAvg	21.5	1.5	1.1	1.3	1.2	1.5	2.2	2.0	2.6	2.0	2.2	1.9	1.9		
(15) Precipitation	PrecMax	30.7	3.5	2.4	2.6	3.1	2.5	4.9	5.0	5.6	5.8	5.2	4.0	5.1		
	PrecMin	13.1	0.1	0.3	0.2	0.2	0.5	0.0	0.3	0.9	0.9	0.5	0.4	0.3		
	PrecStd	3.9	0.8	0.5	0.7	0.7	0.6	1.3	1.1	1.4	1.2	1.1	0.8	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.4	44.4	48.8	60.0	68.4	73.4	78.5	77.7	71.0	62.1	54.6	48.8		
		MCWB	44.6	42.6	45.1	54.1	61.6	66.8	68.2	68.9	63.9	57.7	52.9	47.2		
	2%	DB	44.3	42.7	45.9	55.8	64.3	69.7	74.4	74.7	67.8	59.1	52.4	47.3		
		MCWB	42.6	41.3	43.2	50.8	59.1	63.5	67.3	68.0	62.2	56.5	50.8	45.8		
	5%	DB	42.9	41.6	44.2	52.8	61.4	67.0	72.1	72.2	65.4	57.5	51.0	46.1		
		MCWB	41.5	40.5	42.1	49.1	57.1	61.8	66.2	66.6	61.2	55.1	49.3	44.7		
	10%	DB	41.3	40.4	42.9	50.6	59.2	64.9	70.0	69.9	63.6	56.2	49.4	44.9		
		MCWB	40.1	39.5	41.1	47.6	55.5	60.4	65.0	65.0	59.8	54.1	47.8	43.7		
	0.4%	WB	45.0	43.0	45.7	55.1	62.7	67.6	71.0	71.8	65.3	58.8	53.3	47.8		
		MCDB	45.9	43.7	47.8	59.0	67.1	71.9	75.0	75.3	69.0	60.5	54.1	48.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	43.1	41.7	43.9	51.4	60.0	64.7	68.7	69.4	63.3	56.8	51.1	46.3		
		MCDB	44.0	42.4	45.4	53.9	63.3	68.3	72.7	73.1	66.4	58.5	52.1	47.0		
	5%	WB	41.6	40.6	42.6	49.5	57.7	62.8	67.1	67.3	61.8	55.6	49.6	45.0		
		MCDB	42.6	41.3	43.9	51.8	60.5	65.9	70.9	71.0	64.8	57.1	50.8	45.8		
	10%	WB	40.3	39.6	41.3	47.7	56.1	61.1	65.7	65.6	60.3	54.5	48.0	43.8		
		MCDB	41.2	40.3	42.5	49.9	58.7	64.1	69.4	69.3	63.0	56.0	49.2	44.7		
	5% DB	MDBR	4.0	4.0	5.0	6.6	7.4	7.4	7.3	6.7	5.3	4.3	3.8	3.9		
		MCDBR	4.3	4.5	6.4	10.1	10.3	10.5	10.0	9.5	8.0	5.5	4.1	3.8		
		MCWBR	4.5	3.9	4.8	6.6	6.9	7.4	6.3	5.9	5.3	4.5	4.3	4.1		
		MCWBR	4.1	4.2	6.2	8.7	9.2	9.7	9.0	8.5	7.4	5.3	4.0	3.7		
(35) Mean Daily Temperature Range	5% WB	MCWBR	4.3	4.0	4.9	6.3	6.7	7.3	6.3	6.0	5.5	4.7	4.2	4.1		
	Clear-Sky Solar Irradiance	taub	0.315	0.338	0.359	0.394	0.388	0.380	0.402	0.402	0.374	0.355	0.335	0.303		
		taud	2.285	2.320	2.326	2.282	2.371	2.422	2.375	2.380	2.437	2.417	2.326	2.254		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	185	229	254	261	271	274	265	256	248	221	175	157		
		Edh at Noon	17	24	30	36	35	34	35	33	27	22	16	14		
(44) All-Sky Solar Radiation	RadStd		136	322	779	1304	1695	1829	1694	1344	945	465	177	98		
			25	54	72	152	154	137	177	96	94	59	29	11		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (71 neighbors)	+0.70	N/A	N/A	N/A	N/A	N/A	-288	N/A	N/A	

Nomenclature: See separate page